

S. Sandra Bae

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d-rep.org

RESEARCH MISSION

To establish principles for how data representations should be designed and engineered for the reasoning they are intended to support, across digital, physical, and immersive media.

Specialization: Human-Computer Interaction, Data Visualization
Research interests: Data Physicalization, Computational Design & Fabrication, Optimization, Perception & Cognition, Multimodal Interaction

Last Updated (Sept 2026)

EDUCATION

- 2020 - 2025

University of Colorado, Boulder
Ph.D. in Creative Technology & Design
Thesis: Advancing Interactive Physicalizations
- 2018 - 2020

University of California, Davis
M.S. in Computer Science
- 2014 - 2018

University of California, Davis
B.A. in Human-Computer Interaction
Minor in Education

EMPLOYMENT

- FALL 2025 -
PRESENT

University of Arizona | Tucson, AZ
Assistant Professor, Department of Computer Science
Faculty Affiliate, Cognitive Science
- WINTER 2024 -
SUMMER 2025

National Renewable Energy Laboratory (NREL) | Golden, CO
Graduate Research Assistant
Research with Kenny Gruchalla, Olga Doronina, and Ryan King; collaboration with NIST (Zachary Grey).
- FALL 2020 -
SUMMER 2025

University of Colorado Boulder | Boulder, CO
Graduate Research Assistant
Research with Danielle Albers Szafir, Ellen Yi-Luen Do, and Michael L. Rivera.
- SUMMER 2020

Stanford University | Palo Alto, CA (Remote)
HCI Research Intern
Research with James Landay and Elizabeth Murnane.
- SUMMER 2019 -
SUMMER 2020

NASA Jet Propulsion Laboratory | Pasadena, CA
University Researcher; Data Visualization / HCI Research Intern
Research with Federico Rossi, Scott Davidoff, Joshua Vander Hook, Hillary Mushkin, Santiago Lambledya, and Maggie Hendrie.
NASA JPL Master's Thesis Educational Fellow.

SUMMER 2019 - **University of California, Davis** | Davis, CA
 SPRING 2020 *Undergraduate / Graduate Student Research Assistant*
 Research with Kwan-Liu Ma.

AWARDS AND DISTINCTIONS

2026 Best Poster Honorable Mention (ACM TEI26)
 2025 **Best Paper (ACM CHI25) ***
 2024 **APUP Researcher at the National Renewable Energy Laboratory (\$230,000) ***
 2024 KASF-KIA Scholarship (\$2,500) (Top 5 recipient for KIA special recognition for excellence in STEM) *
 2024 CU Boulder's Special Recognition Award for Partnership for Sustainability (*Coffee Ground Filament*)
 2023 **EECS Rising Star ***
 2023 **Best Paper Honorable Mention (IEEE VIS23) ***
 2023 Korean American Scholarship Foundation (\$2,000)
 2023 Achievement Reward for College Scientists (\$7,500)
 2023 **Rocky Mountain RepRap Festival, Most Functional/Useful, Sensing Networks ***
 2023 Rocky Mountain RepRap Festival, Most Innovative, *Coffee Ground Filament*
 2022 Achievement Reward for College Scientists (\$7,500)
 2022 Korean American Scholarship Foundation (\$3,000)
 2022 David T. Spalding Graduate Teaching Fund Fellowship (\$1000)
 2021 Korean American Scholarship Foundation (\$2,000)
 2021 CU Boulder Travel Grant (\$500)
 2021 CU Boulder Dean's Engineering Travel Grant (\$400)
 2021 Honorable Mention, IEEE World Haptics Student Design Competition
 2021 CRA-WP Grad Cohort for Women
 2021 Ada Lovelace Fellow, Open Hardware Summit
 2021 Craft Award, ACM TEI Student Design Competition.
 2021 Achievement Reward for College Scientists (\$6,500)
 2019 Richard C. and Joy Dorf Engineering Graduate Fellowship (\$1,000)
 2019 **NASA JPL Master's Thesis Educational Fellow (\$30,000) ***
 2018 NSF's Preparing Engineering Graduate Students for the 21st Century Fellow (\$10,000).
 2018 Honorable Mention Visual Storytelling Award, IEEE Pacific Visualization Symposium.
 2017 Beneath (CONNECT EXPO 2017) - Judge's Honorable Mention.
 2017 UC Davis Organizational Research Expo.
 2017 Better Together - AIGA and IDSA.

REFEREED PUBLICATIONS

I publish across computer science and scientific venues. In computer science, conferences are the primary venues (~25% acceptance rates). My work appears in ACM CHI Conference on Human Factors in Computing Systems (CHI), the ACM conference on Designing Interactive Systems (DIS), IEEE Visualization Conference (IEEE VIS), and IEEE International Conference on Robotics and Automation (IEEE ICRA). My interdisciplinary work also appears in peer-reviewed scientific journals spanning aerospace and the life sciences.

UNDER REVIEW

- R1 F. M. Sayegh, C. Scott, K. Ochoa, S. S. Bae, and M. D. Johnson. "An Interactive Publication Framework Demonstrates Carbon Source-Dependent Growth Heterogeneity Across *Streptococcus pneumoniae* Serotypes". In: *mSphere* (2026).

JOURNAL ARTICLES

- J1 S. S. Bae, K. Cave, C. Görg, P. Rosen, D. A. Szafir, and C. X. Bearfield. "Bridging Network Science and Vision Science: Mapping Perceptual Mechanisms to Network Visualization Tasks". In: *IEEE Transactions on Visualization and Computer Graphics* (2025).

- J2 S. S. Bae, T. Fujiwara, A. Ynnerman, E. Y.-L. Do, M. Rivera, and D. A. Szafrir. "A Computational Design Pipeline to Fabricate Sensing Network Physicalizations". In: *IEEE Transactions on Visualization and Computer Graphics (also proc. IEEE VIS 2023)*. Melbourne, Australia, 2023, Acceptance rate: 24.7% (133/539)
* Best Paper Honorable Mention (Top 3%).
- J3 S. S. Bae, R. Vanukuru, R. Yang, P. Gyory, R. Zhou, E. Y.-L. Do, and D. A. Szafrir. "Cultivating Visualization Literacy for Children Through Curiosity and Play". In: *IEEE Transactions on Visualization and Computer Graphics (also proc. IEEE VIS 2022)*. 2022, Acceptance rate: 26.5% (122/460).
- J4 R. Woollands, F. Rossi, T. S. Vaquero, M. S. Net, S. S. Bae, V. Bickel, and J. V. Hook. "Maximizing Dust Devil Follow-Up Observations on Mars Using Cubesats and On-board Scheduling". In: *Journal of Astronautical Sciences*. 2022.

CONFERENCE ARTICLES

- C1 H. C. van Iterson, T. Lu, S. S. Bae, S. Houben, and R.-H. Liang. "PaperBlox: Reducing Equipment Barriers in TUI Prototyping via Paper Templates and Computer Vision". In: *Proceedings of NordiCHI '26: Proceedings of the 14th Nordic Conference on Human-Computer Interaction*. Vaasa, Finland, 2026, Acceptance rate: 21.8% (110/504).
- C2 K. Ranjan, S. S. Bae, P. Gyory, E. Y.-L. Do, R.-H. Liang, and C. Zheng. "Why (not) ReacTiVision: Emerging Challenges and Opportunities for Building Tangible User Interfaces with Computer Vision Toolkits". In: *Proceedings of the Twentieth International Conference on Tangible, Embedded, and Embodied Interaction (TEI)*. Chicago, USA, 2026, Acceptance rate: 22.65% (41/181).
- C3 S. S. Bae*, T. Fujiwara*, D. A. Szafrir, E. Y.-L. Do, and M. L. Rivera. "Computational Design and Single-Wire Sensing of 3D Printed Objects with Integrated Capacitive Touchpoints". In: *Proceedings of the 10th ACM Symposium on Computational Fabrication*. Boston, USA, 2025, (*equally contributed).
- C4 S. S. Bae*, T. Fujiwara*, C. Tseng*, and D. A. Szafrir. "Uncovering How Scatterplot Features Skew Visual Class Separation". In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. Yokohama, Japan, 2025, Acceptance rate: 25.1% (1249/5020).
- C5 C. Kohlbrenner, C. Esobedo, S. S. Bae, A. Dickhans, and A. Roncone. "GenTact Toolbox: A Computational Design Pipeline to Procedurally Generate Context-Driven 3D Printed Large-Area Tactile Skins". In: *Proceedings of the 2025 IEEE International Conference on Robotics and Automation*. Atlanta, Georgia, 2025, Acceptance rate: 38.67% (1606/4153).
- C6 R. Sorenson-Graff, S. S. Bae, and J. Wirfs-Brock. "Integrating Annotations for Sonifications and Physicalizations". In: *Proceedings of the IEEE VIS 2024*. St. Pete Beach, Florida, 2024, Acceptance rate: 31.9% (66/207).
- C7 X. Wen, S. S. Bae, and M. L. Rivera. "Enabling Recycling of Multi-Material 3D Printed Objects through Computational Design and Disassembly by Dissolution". In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. Yokohama, Japan, 2024, Acceptance rate: 25.1% (1249/5020)
* Best Paper (Top 1%).
- C8 P. Gyory, S. S. Bae, R. Yang, E. Y.-L. Do, and C. Zheng. "Marking Material Interactions with Computer Vision". In: *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. Hamburg, Germany, 2023, Acceptance rate: 28.39% (880/3182).
- C9 M. Rivera, S. S. Bae, and S. Hudson. "Exploring Spent Coffee Grounds as a Sustainable Material for Prototyping with 3D Printing". In: *Proceedings of the 2023 on Designing Interactive Systems Conference*. Pittsburgh, PA, USA, 2023, Acceptance rate: 24.1% (174/726).
- C10 S. S. Bae, D. A. Szafrir, and E. Y.-L. Do. "Exploring the Benefits and Challenges of Data Physicalization". In: *Proceedings of the Fourth European Tangible Interaction Studio (ETIS' 22)*. Toulouse, France, 2022, Acceptance rate: 88% (15/17).
- C11 S. S. Bae, C. Zheng, M. E. West, E. Y.-L. Do, S. Huron, and D. A. Szafrir. "Making Data Tangible: A Cross-disciplinary Design Space for Data Physicalization". In: *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. New Orleans, Louisiana, 2022, Acceptance rate: 23% (4590/19964).
- C12 S. S. Bae, F. Rossi, J. V. Hook, S. Davidoff, and K.-L. Ma. "A Visual Analytics Approach to Debugging Cooperative, Autonomous Multi-Robot Systems' Worldviews". In: *Proceedings of the IEEE Visual Analytics Science and Technology (VAST) 2020*. Salt Lake City, Utah (Virtual), 2020, Acceptance rate: 24.8% (52/210).
- C13 S. S. Bae*, O.-H. Kwon*, S. Chandrasegaran, and K.-L. Ma. "Spinneret: Aiding Creative Ideation through Non-Obvious Concept Associations". In: *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. Honolulu, Hawaii (Virtual), 2020, (*equally contributed), Acceptance rate: 24.31% (760/3126).

DEMOS & POSTERS

- D1 R.-H. Liang, S. Houben, K. Ranjan, S. S. Bae, P. Gyory, E. Y.-L. Do, and C. Zheng. “ArUcoTUI: Software Toolkit for Prototyping Tangible Interactions on Portable Flat-Panel Displays with OpenCV”. In: *Proceedings of the Twentieth International Conference on Tangible, Embedded, and Embodied Interaction (TEI)*. Chicago, USA, 2026.
- D2 S. S. Bae*, T. Fujiwara*, D. A. Szafir, E. Y.-L. Do, and M. L. Rivera. “Demonstration of Computational Design and Single-Wire Sensing of 3D Printed Objects with Integrated Capacitive Touchpoints”. In: *Proceedings of the 10th ACM Symposium on Computational Fabrication*. Boston, USA, 2025, (*equally contributed).
- D3 S.-Y. Ma, S. S. Bae, E. Y.-L. Do, T. Yeh, and E. Bradley. “From Flat Screens to Immersive Space: Rethinking Multimodal Learning of Complex Systems”. In: *Proceedings of the Eleventh Taiwan Human-Computer Interaction Symposium (TAICHI '25)*. Hsinchu, Taiwan, 2025.
- D4 X. Wang, H. C. van Iterson, S. S. Bae, and R.-H. Liang. “Reflecting on Solo Dining Behavior with Annotated Data Physicalization”. In: *Proceedings of the Eighteenth International Conference on Tangible Embedded and Embodied Interaction (TEI '25)* (2025).
- D5 S. S. Bae, T. Fujiwara, D. A. Szafir, E. Y.-L. Do, and M. L. Rivera. “3D printed multi-point capacitive touch sensor”. In: *Symposium on Computational Fabrication*. SCF '24. Aarhus, Denmark: Association for Computing Machinery, 2024.
- D6 E. Johnson, S. S. Bae, and E. Y.-L. Do. “Supporting Data Visualization Literacy through Embodied Interactions”. In: *Proceedings of the Fifteenth International Conference on Creativity & Cognition (C&C '22)*. Gathertown (Remote): Association for Computing Machinery, 2023.
- D7 S. S. Bae. “Towards a Deeper Understanding of Data and Materiality”. In: *Proceedings of the Fourteenth International Conference on Creativity & Cognition (C&C '22)*. Venice, Italy: Association for Computing Machinery, 2022.
- D8 S. S. Bae, R. Yang, P. Gyory, J. Uhr, D. A. Szafir, and E. Y.-L. Do. “Touching Information with DIY Paper Charts & AR Markers”. In: *Interaction Design and Children*. IDC '21. Athens, Greece (Virtual): Association for Computing Machinery, 2021, Acceptance rate: 27% (136/501).
- D9 S. S. Bae* and M. E. West*. “Cyborg Crafts: Second SKIN (Soft Keen INteraction)”. In: *Proceedings of the Fifteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '21)*. Salzburg, Austria (Virtual): Association for Computing Machinery, 2021, (*equally contributed).

TECH REPORTS

- T1 R. Woollands, F. Rossi, T. S. Vaquero, M. S. Net, S. S. Bae, V. Bickel, and J. V. Hook. “Maximizing Dust Devil Follow-Up Observations on Mars Using Cubesats and On-board Scheduling”. In: *Proceedings of the 43rd Annual AAS Guidance & Control Conference, Breckenridge, CO*. 2021.

BOOK CHAPTERS

- B1 T. Hopkins, S. S. Bae, J. Uhr, C. Zheng, A. Banic, and E. Y.-L. Do. “User Interfaces in Smart Cities”. In: *Handbook of Smart Cities*. Ed. by J. C. Augusto. Springer International Publishing, 2021.

PATENT AND INDUSTRY PARTNERSHIP

NON-PROVISIONAL PATENT APPLICATION

Bae, S. S, Fujiwara, T., Rivera, M.L., Szafir, D.A., Do., E.Y.D (2023). Systems and Methods for Touching Sensing Based on Resistor-Capacitor Delays. *U.S. Patent Application 63/497,594*, filed April 21, 2023. Pending non-provisional patent application.

PROVISIONAL PATENT APPLICATION

Bae, S. S, Fujiwara, T (2025). Systems and Fabrication Methods for Resistor Designs Through Printing. *UA26-100P*, filed November 16, 2025.

INDUSTRY PARTNERSHIP

Company	Funding	Year
Stratasys	Gift	2024
Protopasta	Sponsored Research	2024, 2025

TALKS

All first-author conference papers listed above were also given as presentations at their respective conferences and are not listed again in this section.

INVITED TALKS AND PRESENTATIONS

Network Perception: For network visualizations on and beyond the screen

Dagstuhl Seminar: Perception-Informed Network Visualization (#26361), Wadern Germany, August 31, 2026

D-REP Lab Research Examples: Computational Fabrication

Sony CSL Kyoto, Japan, July 26, 2026 (hosted by Yusuke Sakai)

Discussions on the Health Design Implementation and AI

University of Arizona, June 8, 2026 (Invited Panelist)

From Pixels to Physicalization: Perception as a Foundation for Visualization Design

University of Arizona, April 23, 2026 (hosted by Jonathon Tullis)

Communicating Complex Data through 3D Data Representation

University of Arizona, April 3, 2026 (hosted by Sara Kobilka)

Communicating Complex Data through 3D Data Representation

University of Arizona Biomedical Engineering Colloquium, February 9, 2026 (hosted by Swarna Garnesh)

Escaping Flatland: Data Physicalization Now, Multi-Modal Agents Next

University of Utah, December 3, 2025 (hosted by Andrew McNutt)

AI, Meet Matter: How AI Generates Forms, Mediates Interactions, and Models Cognition

University of Arizona AI Club, November 12, 2025 (hosted by Zhuolin Yang)

IEEE VIS 2025 Doctoral Colloquium

Panelist, IEEE VIS 2025 Doctoral Colloquium

Escaping Flatland: 3D+ Data Visualizations for New Users, Interactions, and Analytical Environments

TU Wien, October 31, 2025 (hosted by Renata Raidou)

Simple Tools, Creative Outcomes: A Computer Vision-Based Approach

Computer Science Teachers Organization (CSTA-Arizona), September 11, 2025 (hosted by Mihai Surdeanu)

Escaping Flatland: 3D+ Data Visualizations for New Users, Interactions, and Analytical Environments

University of Texas, Dallas (UT Dallas)

University of Arizona

Worcester Polytech Institute (WPI)

Washington University in St. Louis

Tufts University

Purdue University

Lifting Visualizations Beyond Flatland: A Geometric Perspective

National Institute of Science and Technology (NIST), October 1, 2024 (hosted by Matt Whitlock)

Pushing Visualizations Beyond the Desktop with Everyday Devices and Novel Interfaces
University of Iowa, November 17, 2023 (hosted by Juan Pablo Hourcade)

Pushing Visualizations Beyond the Desktop with Everyday Devices and Novel Interfaces
Emory University, November 8, 2023 (hosted by Emily Wall)

Pushing Visualizations Beyond the Desktop with Everyday Devices and Novel Interfaces
Georgia Tech, November 8, 2023 (hosted by Yalong Yang)

Using Network Science to Reimagine Everyday Objects as Future Computers
TTI Vanguard, September 13, 2023 (hosted by Nancy Kleinrock)

Towards “Best of Both Worlds”: Bridging Digital and Physical Representations in Visualization Research
Linköping University, January 9, 2023 (hosted by Miriah Meyer)

Debugging Multi-Robot Autonomous System Anomalies
CU Boulder ATLAS Seminar, October 6, 2020 (hosted by Ellen Yi-Luen Do)

Debugging Multi-Robot Autonomous System Anomalies
NASA-JPL, August 27, 2020 (hosted by Scott Davidoff)

INVITED GUEST LECTURER

Escaping Flatland: Data Physicalization Now, Multi-Modal Agents Next
Seoul National University, *Information Visualization and Visual Analytics*, December 7, 2025 (hosted by Jinwook Seo)

Expanding your Interactive Toolkit: Computer Vision Markers and 3D Printed Capacitive Sensors
Georgia Tech, *Prototyping Interactive Systems*, November 7, 2023 (hosted by Hyunjoo Oh)

Using Computational Fabrication for Data Physicalization
UNC-Chapel Hill, *Information Professionals in the Makerspace*, October 17, 2023 (hosted by Maggie Melo)

Computational Fabrication for Electronics
CU Boulder, *Computational Fabrication*, October 16, 2023 (hosted by Michael L. Rivera)

The Value of Design: Empowering Humans through Human-Centered Thinking
CU Boulder, *Information Visualization*, March 18, 2021 (hosted by Danielle Szafir)

The Value of Design: Empowering Humans through Human-Centered Thinking
UC Davis, *Android Development*, January 14, 2021 (hosted by Nina Amenta)

TEACHING

COURSES TAUGHT

[†] indicates a course developed

Course	Institute	Term(s)	Enrollment
CSC596: Human-Computer Interaction (Graduate) [†]	UArizona	FALL 2026	4
CSC696C: Compute, Fabricate, Interact [†]	UArizona	SPRING 2026	6

PRIOR TEACHING EXPERIENCE

Course	Institute	Term(s)	Enrollment
3D Printing (Brave Behind Bars ¹)	UWashington	FALL 2025	12
Web Development (Brave Behind Bars ¹)	MIT	SUMMER 2024	54
ATLAS3300: Object (Physical Computing)	CU Boulder	SPRING 2021 SPRING 2022	36 37
ECS162: Web Programming	UC Davis	SPRING 2020	122
ECS164: Intro to Human-Computer Interaction	UC Davis	WINTER 2019 WINTER 2020	63 59
ECS163: Information Visualization	UC Davis	SUMMER 2018	23

¹ Brave Behind Bars is a college-accredited introductory computer science and career-readiness program for incarcerated people.
<https://bravebehindbars.org/>

ADVISING & MENTORSHIP

ADVISED STUDENTS

Name	Degree	Year
Ph.D. Students		
Taoseef Aziz	Ph.D. in Computer Science (University of Arizona)	2026 – PRESENT
Aditya Retnanto	Ph.D. in Computer Science (University of Arizona)	2026 – PRESENT
Yanyan Dong	Ph.D. in Information Science (University of Arizona) * co-advised with Steve Bethard	2025 – PRESENT
Undergraduate Students		
Jordan Orvik	B.S. in Computer Science (University of Arizona)	2025 – 2026

RESEARCH MENTORSHIP

Name	Degree	Year
Ph.D. Students		
Krithik Ranjan	Ph.D. in Creative, Tech. & Design (CU Boulder)	2024 – 2025
Xin Wen	Ph.D. in Creative, Tech. & Design (CU Boulder)	2024
Chin Tseng	Ph.D. in Computer Science (UNC-Chapel Hill)	2023 – 2024
Peter Gyory	Ph.D. in Creative, Tech. & Design (CU Boulder)	2022 – 2023
Master's Students		
Quintan Gerhardstein	M.S. in Creative, Tech. & Design (CU Boulder)	2024
Ada Zhao	M.S. in Creative, Tech. & Design (CU Boulder)	2024
Carson Kohlbrenner	M.S. in Aerospace Engineering (CU Boulder)	2023–2024
Ashleigh Thomas	M.S. in Computer Science (UC Davis)	2019
Undergraduate Students		
Shreyans Daga	B.S. in Computer Science Engineering (University of Arizona)	2026 – PRESENT
Gwendolyn Alviar	B.S. in Mechanical Engineering (University of Arizona)	2025 – PRESENT
William Bettis	B.A. in Design Arts & Practices (University of Arizona)	2025 – PRESENT
Noah Liska	B.S. in Computer Science (CU Boulder)	2023–2024
Caleb Kumar	B.S. in Computer Science (CU Boulder)	2023–2024
Rhys Sorenson-Graff	B.A. in Computer Science (Whitman College)	2023–2024
Eloise Yalovitser	B.F.A in Design & Tech. (Parsons School of Design)	2023

Antonio (Zhixing) Li	B.S. in Mechanical Engineering (CU Boulder)	2023
Elise Johnson	B.S. in Mechanical Engineering (CU Boulder)	2021–2022
Vy Thai	B.S. in Computer Science (Stanford University)	2020
Lovpret Kaur	B.S. in Computer Science (UC Davis)	2019

THESIS & DISSERTATION SUPERVISION

		PhD	MS	BS
Advisor	Completed	0	0	1
	Current	3	0	0
Committee Member	Completed	0	1	–
	Current	1	0	–

PhD Dissertations—Committee

Name	Title	(Dept.) Year
Tanner A. Finken	Discrete Vector Field Improvement, Supporting Fast Generation, Provable Correctness, and Adaptable Simplification	2026

MS Theses—Committee

Name	Title	(Dept.) Year
Md Afridi Hasan	A Perturbation-driven Framework for Sensitivity Analysis of Dimensionality Reduction	2026

BS Theses—Advisor

Name	Title	(Dept.) Year
Jordan M. Orvik	Mediating the Follow: A Longitudinal Study on Reflective Practice and its Effect on Parasocial Relationships on Instagram	2026

ACADEMIC SERVICE TO PROFESSIONAL COMMUNITY

Associate Chair

ACM Conference on Human Factors in Computing Systems (CHI) 2027

Program Committee

ACM Creativity & Cognition (C&C) 2021

ACM Tangible and Embedded Interaction (TEI) 2024

IEEE VIS 2026

IEEE Pacific Visualization Conference (Visualization Notes) 2025

EG/VGTC EuroVis (Short Paper) 2026

Organizing Committee

IEEE VIS 2026, 2027 (Inclusivity Chair)

IEEE VIS 2022, 2024 BELIV Workshop (Social Media and Web Chair)

IEEE VIS 2026 BELIV Workshop

Invited Conference Reviewer

ACM Symposium on User Interface Software and Technology (UIST) 2021, 2026

ACM Interaction Design and Children (IDC) 2023

ACM Designing Interactive Systems (DIS) 2021–2024

ACM Conference on Human Factors in Computing Systems (CHI) 2021–2026

ACM Tangible and Embedded Interaction (TEI) 2022–2027

ACM Creativity & Cognition (C&C) 2021–2022

EG/VGTC EuroVis 2022, 2026

NordiCHI 2024

SIGGRAPH Asia Technical Paper 2024

SIGGRAPH Technical Paper 2025, 2026

IEEE VR 2025

IEEE Computer Graphics & Applications 2026

IEEE Robotics and Automation Letters 2026

Invited Journal Reviewer

IEEE Transactions on Visualization and Computer Graphics (TVCG) 2022–2026

International Journal of Child-Computer Interaction 2022

Behaviour & Information Technology 2023, 2024

IS&T Journal of Perceptual Imaging 2024, 2025

International Journal of Human-Computer Studies 2025

Invited Book Chapter Reviewer

“Visualization Psychology” published by Springer Nature. Editors: Danielle Albers Szafir, Rita Borgo, Min Chen, Darren J. Edwards, Brian Fisher, & Lace Padilla

Grant Reviewing

Panelist, NSF IIS-Human-Centered Computing (HCC)

Special Recognition as a Reviewer

2023 ACM Tangible and Embedded Interaction (TEI)

2023 ACM Conference on Human Factors in Computing Systems (CHI) x2

2024 ACM Conference on Human Factors in Computing Systems (CHI) x2

2026 ACM Conference on Human Factors in Computing Systems (CHI)

Other Conference Organizational Roles

Faculty Panelist, IEEE VIS 2025 Doctoral Colloquium

Session Chair, IEEE VIS 2025 Full Paper Presentation

ACADEMIC SERVICE TO UNIVERSITY

Departmental Service

Graduate Admissions Committee, Department of Computer Science, 2025–2027

Seminar Chair, Department of Computer Science, 2025–2027

Ph.D. Open House Organizer 2021, ATLAS Institute, CU Boulder

Faculty Candidate Student Host 2021, 2024 ATLAS Institute, CU Boulder
Ph.D. Graduate Application Support Program Founder 2022, 2023, ATLAS Institute, CU Boulder

DESIGN EXHIBITIONS

2019 **Sit On Data** | Imagining America's 20th Anniversary National Gathering

A data-driven parametric-bench where users can feel the data by sitting. Developing using Python, CNC routing, Laser cutting.

2017 **BENEATH** | SOFA Chicago 2017

An immersive installation conveying California's land subsidence. Presented at SOFA Chicago 2017. Built motion graphics and projection using Processing.

2017 **OneClimate: Prototyping Climate Change** | Exploratorium Museum (SF)

A multisensory exhibition conveying the impact of climate change sponsored by the John Muir Institute of the Environment. Featured at the Exploratorium in San Francisco for the OneClimate event, AIGA & ISAD - San Francisco, and IA's 2017 National Gathering. Designed the scientific narrative and constructed the exhibit using woodshop skills.

SELECTED MEDIA COVERAGE

2023 **Yahoo Finance** | First-of-its-kind study combines common household waste product with 3D printing — here are the results

<https://finance.yahoo.com/news/first-kind-study-combines-common>

2023 **Ars Technica** | Don't throw out those used coffee grounds—use them for 3D printing instead

<https://arstechnica.com/science/2023/09/dont-throw-out-those-used-coffee-grounds-use-them-for-3d-printing-instead/>

2023 **TechBriefs** | Using Coffee to Reduce Waste from 3D Printing

<https://www.techbriefs.com/component/content/article/tb/stories/blog/49103>

2023 **Daily camera (Boulder)** | CU Boulder professor creates 3D printing paste with coffee grounds

<https://www.dailycamera.com/2023/09/20/cu-boulder-professor-creates-3d-printing-paste-with-coffee-grounds/>

2023 **CNC Kitchen (Youtube)** | 3D Printing Resistors, Fibers & Coffee - Utility Research Lab

https://youtu.be/Oeqvo2c28_c

2023 **Make Magazine** | Rocky Mountain RepRap Fest: All The Cool 3D Printing Stuff

<https://makezine.com/article/digital-fabrication/3d-printing-workshop/3d-printing-festival-loveland-colorado/>

2023 **Meltzone Podcast** | Rocky Mountain RepRap Festival & Prusa MK4

<https://youtu.be/NkgOXFn55cM?t=1041>

2023 **CU Boulder** | ATLAS Innovators Win Big at RepRap Festival

<https://www.colorado.edu/atlas/2023/05/08/atlas-innovators-win-big-reprap-festival>

2023 **CU Boulder** | New seed grants from the Engineering Education and AI-Augmented Learning research theme will support range of topics

<https://www.colorado.edu/irt/engineering-education-ai/2023/05/19/new-seed-grants-engineering-education-and-ai-augmented-learning-research-theme-will>

2021 **CU Boulder** | Sandra Bae receives ARCS and KASF scholarship

<https://www.colorado.edu/atlas/2021/09/08/sandra-bae-receives-arcs-and-kasf-scholarships>

2017 **CONNECT EXPO** | SOFA CHICAGO 2017 Exhibition

Printing Press: Chubb (pg. 10-11).

2017 **AIGA and IDSA** | Better Together Design Exhibition

Printing Press: AIGA (pg. 8-9).

2017 UC Davis | 'OneClimate' a Call to Arms
<https://www.ucdavis.edu/news/oneclimate-call-arms>